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Continuous motion tracking for accurate and efficient color vision assessment

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Abstract
The assessment of color vision is crucial in both fundamental visual research and clinical diagnosis. However, existing tools for color vision assessment are limited by various factors. This study introduces a novel, efficient method for color vision assessment, which is based on a continuous motion tracking task and a Kalman filter model. The effectiveness of this new method was evaluated by assessing the color vision of both color-deficient observers and normal controls. The results from both a small sample ($N=29$, Experiment 1) and a large sample ($N=171$, Experiment 2) showed that color-deficient observers could be perfectly identified within 20 s using the tracking performance. We also compared the new method with a traditional psychophysical detection task to examine the consistency of perceptual noise estimation between the two methods, and the results showed a moderate correlation (Pearson's $r=.59\sim.64$). The results also demonstrated that the new method could measure individuals' contrast response functions of both red–green and blue–yellow colors (e.g., the L–M and S–(L + M) axes in DKL color space) in just a few minutes, showing much higher efficiency than traditional methods. All the findings from this study indicate that the continuous motion tracking method is a promising tool for both rapid screening of color vision deficiencies and fundamental research on color vision.

Keywords Continuous psychophysics · Color vision · Kalman filter · Motion tracking

Introduction

Color vision is a fundamental function of the human vision system, and researchers have developed various tools for its assessment. Certain tools, like the Ishihara plates, employ pseudoisochromatic plates to quickly screen for color vision deficiencies, but they cannot provide precise assessments of an individual's sensitivity to specific colors (Rodriguez-Carmona et al., 2012). Other tools use traditional psychophysical procedures to accurately measure individuals' sensitivity to color contrasts, but they are time-consuming and primarily used in the laboratory. There is currently no well-established tool available that can provide both quick and precise assessments of individuals' sensitivity to specific color contrasts. In this study, we introduce a new method using a continuous motion tracking task and a Kalman filter model, which can achieve precise measurement of color sensitivity in a short amount of time. Additionally, we attempted to use this new method to construct contrast-response functions (CRFs) that provide a more comprehensive description of observers' color vision, as opposed to a single threshold index from traditional psychophysical procedures.

The existing tools for color vision assessment

Human color vision, as one of the most fundamental visual functions, enables human observers to identify over two million colors (Jacobs, 2009). Given its importance in people's interaction with environments, researchers have long been developing tools for color vision assessment (e.g. Pickgord, 1944; Rabin et al., 2010) in order to (1) distinguish

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individuals with color vision deficiency from those with normal color vision and (2) comprehensively evaluate individuals' color vision.

Color vision deficiency (CVD) refers to anomalies in sensation and perception of colors, which can be congenital or acquired and has a higher prevalence in males (8%) than in females (0.5%) due to dysfunctions caused by X chromosome genetics (Simunovic, 2010). Since normal color vision requires three types of cone photoreceptors on the retina, with each sensitive to short wavelengths (S cone), medium wavelengths (M cone), and long wavelengths (L cone), dysfunctions in certain cones lead to different types of CVD. Loss of one cone type leads to dichromacy, and a shift in wavelength sensitivity of the cone leads to anomalous trichromacy (Merbs & Nathans, 1992). Depending on the affected cones, dichromacy includes protanopia (lack L), deuteranopia (lack M), and tritanopia (lack S, very rare). Anomalous trichromacy includes protanomaly (shift in wavelength sensitivity of L), deuteranomaly (shift in wavelength sensitivity of M), and tritanomaly (shift in wavelength sensitivity of S, very rare).

There are various tools for CVD assessment, including tests using anomaloscopes, which are generally considered the gold standard of CVD classification (Zarazaga et al., 2019). Nevertheless, the most frequently used tools are pseudoisochromatic plate tests for convenience in clinical practice. Pseudoisochromatic plate tests such as the Ishihara test (Pickgord, 1944), the Richmond HHR (Hardy–Rand–Rittler) tests (Cole et al., 2006; Hardy et al., 1947), and Cambridge color test (Mollon & Regan, 2000) have been widely utilized in medical and vision care settings due to their simplicity and effectiveness. Standardized with carefully designed graphics, the test provides objective results (Barbur & Rodriguez-Carmona, 2012; Jurasevska et al., 2014). However, despite their utility in classifying different types and severity of CVD (Cole et al., 2006; Mollon & Regan, 2000), the applicability of pseudoisochromatic plate tests is limited due to the lack of detailed information on individuals' sensitivity to colors along different color axes.

Color vision is a multidimensional structure that includes continua of sensitivity to specific color contrasts, which are highly complex and vary significantly across individuals, including both people with CVD and those with normal color vision. Thus, merely screening types and severity of CVD does not offer a comprehensive evaluation of individuals' color vision. For this purpose, traditional psychophysical procedures with forced-choice tasks are the preferred and most commonly used tool to measure thresholds for color contrasts in fundamental research on color vision, as these methods can provide precise descriptions of individuals' color vision (Pelli & Bex, 2013). For example, many researchers used discrimination tasks to precisely measure individuals' contrast sensitivity and further construct contrast sensitivity functions (Anstis & Cavanagh, 1983; Hansen & Gegenfurtner, 2013; Krauskopf & Gegenfurtner, 1992), which are widely considered the most comprehensive and informative measure of color vision (see Pelli & Bex, 2013).

Despite the advantages in precision, comprehensiveness, and informativeness, traditional psychophysical methods are nevertheless not often used in general visual research and are rarely used in clinical screening for visual deficiency because of their immense time demands. Since the establishment of the psychophysical method by Fechner (1860), data collection in psychophysical experiments has been inefficient and much more time-consuming than other behavioral experiments. For example, determining a threshold for an individual at a specific color contrast tends to require hundreds, if not thousands, of binary judgements, each of which may take several seconds but yields no more than 1 bit of information. Although the design might be shortened with the use of adaptive procedures and Bayesian algorithms (Lesmes et al., 2006, 2010; Vul et al., 2010), 80–100 trials are still required for a stable measure of a single threshold/sensitivity, making it unrealistic to adopt these procedures in most practical cases.

In short, there is not yet a tool for color vision assessment which can provide both quick screening of CVD and comprehensive measures of individuals' color vision. The current study sought to resolve this issue by introducing a motion tracking task and a Bayesian framework, both of which are further discussed in the following subsection.

Vision assessment by motion tracking tasks

Several recent studies have attempted to increase the efficiency of psychophysical measurements in vision research by using innovative tasks (e.g., (Bonnen et al., 2015, 2017; Dorr et al., 2017; Mooney et al., 2018). One simple but effective solution is the continuous recording of participants' responses to dynamic stimulus, which enables the collection of a large amount of information in a relatively short amount of time. Bonnen et al. (2015) first introduced a dynamic motion tracking task to evaluate an individual's visual sensitivity, such as luminance contrast. Multiple studies have now adopted motion tracking tasks to investigate individuals' visual acuity (e.g., Bonnen et al., 2015, 2017; Dorr et al., 2017; Mooney et al., 2018), stereopsis (Bonnen et al., 2017), and numerosity representation (Ambrosi et al., 2021). In these studies, each instance of stimulus change during the task was treated as a "mini-trial," and response variations related to the stimulus change were interpreted as a reflection of visual processing. Based on this approach, one can collect data from thousands of mini-trials within a single minute.

However, a significant challenge in the motion tracking task is how to extract an estimate from performance in a tracking task so that the estimate is analogous to threshold measured from a traditional psychophysical procedure. Different from the traditional methods which use trial-based forced-choice tasks, response variations among participants in motion tracking are not simply from perceptual noise. Researchers (Baddeley, 2003; Bonnen et al., 2015) have pointed out that manual tracking by people can be explained by the optimal control theory of tracking in engineering. Observers can be treated as Bayesian optimal trackers, with their tracking performance described using the Kalman filter model (Kalman, 1960). As Fig. 1 shows, tracking performance can be modeled by a simple dynamic linear system, in which tracking performance at a specific moment is determined by an optimal combination of two factors, sensory input in effect at the same moment and tracking performance at the previous moment. Note that the performance at the previous moment includes a weighted aggregation of sensory input across all previous moments, and the dynamic weighting of previous moments can stabilize observers' motor control when sensory input is poor and, to a certain extent, account for correlations between tracking performance at sequential moments (Bonnen et al., 2015; Burge et al., 2008; Körding et al., 2004). More importantly, this model provides a theoretical framework for reversing the visuomotor process, by which noise of sensory input can be separated and independently estimated. This estimated noise is theoretically analogous to "threshold" defined in traditional psychophysical methods and can be used to evaluate individuals' visual functions. Bonnen et al. (2015) first used this framework to extract individuals' sensory noise to displacement at different luminance contrasts. Other studies have also used this model, with dynamic motion tracking tasks, to evaluate various vision-related functions (e.g., Bonnen et al., 2015, 2017; Dorr et al., 2017; Mooney et al., 2018). Straub and Rothkopf (2022) recently published a modified model which introduces a more complex structure with components related to decision-making, so as to minimize the differences in sensitivity estimation between traditional tasks and visuomotor tasks. Enlightened by these previous findings, the current study managed to use the same framework to develop a new method in color vision assessment specifically for screening CVD.

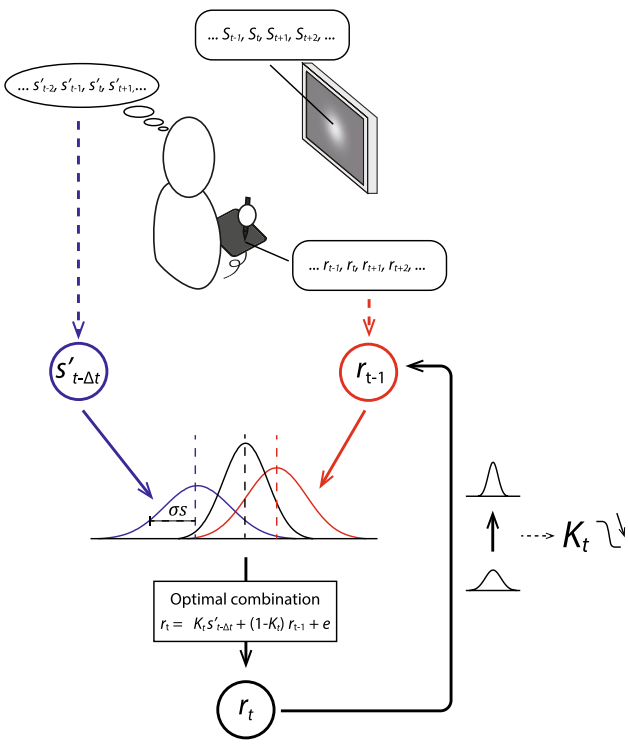

Fig. 1 The dynamic linear system model for the participants' tracking performance. The tracking performance at a given moment (r_t) is determined by a dynamic optimal combination of delayed sensory input ($s'_{t-\Delta t}$) and performance at the previous moment (r_{t-1}). This combination follows a Kalman filter process (for details, please see Appendix A), reflected by the Kalman coefficient K, which is supposed to be relatively high when the participant is more dependent on sensory input compared to performance at the previous moment

The present study

In this study, we aimed to develop a new and efficient method for assessing color vision based on a motion tracking task. The method involved having participants, both with and without CVD, track a moving patch defined by color contrasts in the L–M or S axis of the DKL [Derrington–Krauskopf–Lennie] color space using a digital pen and a trackpad. The tracking performance of the individual participants was analyzed using a Kalman filter model, which allowed us to extract and parameterize observers' sensory noise from their tracking performance. Two experiments were included in this study. Experiment 1 was designed to test the possibility of distinguishing between people with CVD ($N = 14$) and those with normal color vision ($N = 15$) and assessing individual participants' color vision with a motion tracking task. In Experiment 2, we further tested the new method using a larger random sample ($N = 171$) and a short tracking duration (20 s).

As a brief view of the results, screening based on this estimation of sensory noise was perfectly consistent with the results of the Ishihara test, and the estimation was moderately correlated with the estimated thresholds measured by a traditional psychophysical task. These findings confirm the effectiveness and efficiency of the new method in assessing color vision.

Experiment 1

Method

Participants

Fourteen individuals with CVD (13 male, 1 female; average age: 21.1 years; age range: 18–26 years) and 15 individuals with normal color vision (7 male, 8 female; average age: 23.4 years; age range: 18–26 years) participated in the experiment. All participants were recruited from two universities in Shanghai, PR China. The CVD group was recruited through online advertising specifically seeking participants who were color-deficient or who had failed color screenings by the Ishihara test. Seven observers with CVD were tested by an anomaloscope (Oculus Optikgeräte GmbH, Germany), and the results showed that three exhibited protanomaly and four exhibited deuteranomaly. Participants were compensated for their participation. They signed written informed consent forms in accordance with the Declaration of Helsinki. The experimental procedure was approved by the ethics committee of Shanghai University of Sport (no. 102772019RT009).

Apparatus and stimuli

The tasks were coded using the Psychophysics Toolbox (Brainard, 1997; Pelli, 1996) in MATLAB (MathWorks, Natick, MA, USA). The experiment was conducted using a laptop (Acer Predator Helios 300) at a refresh rate of 60 Hz. The motion tracking equipment used in the experiment was the Wacom Intuos CTL-6100 trackpad with a digital pen connected to the laptop. The participant used the digital pen at the trackpad to control the cursor on the monitor, and kept tracking the target stimulus during the experiment.

The monitor has a resolution of 2560 × 1440 pixels. At a viewing distance of 60 cm, the screen extends to 32.0° × 18.0° in visual angle. The monitor was color-calibrated with a spectrophotometer (i1Basic Pro 2, X-Rite, USA). The primaries of our monitor have *xyY* coordinates of red [0.68, 0.32, 25.6 cd/m^2], green [0.27, 0.69, 85.6 cd/m^2], and blue [0.15, 0.061, 10.1 cd/m^2]. The colors of stimuli were specified with the Derrington–Krauskopf–Lennie (DKL) color space (Derrington et al., 1984; Krauskopf et al., 1982). For a detailed description of our DKL color space, please see the appendix of Hansen and Gegenfurtner (2013). Since the DKL color space is device-dependent, we converted the endpoints of the L–M and S axes into *xy* coordinates and cone-contrasts (Brainard, 1997; Noorlander et al., 1981; Smith & Pokorny, 1975). Table 1 shows how endpoints in *xy* coordinates and cone-contrast match in this study.

Table 1 Endpoints of L–M and S axes in *xy* coordinates and cone-contrast

	xy Coordinates	Cone-contrast
L–M	[0.38, 0.29]	[8%, – 16%, 0.3%]
–L + M	[0.21, 0.37]	[– 8%, 16%, – 0.3%]
S – (L + M)	[0.40, 0.57]	[0%, 0%, 95%]
– S + (L + M)	[0.27, 0.23]	[0%, 0%, – 95%]

Figure 2 shows sample stimuli with a red–green target (Fig. 2A) or a blue–yellow target (Fig. 2B). The gray background covers 1000 × 1000 pixels (12.5° × 12.5°), with each small checker being 20 × 20 pixels (0.25° × 0.25°). The color target is isoluminant to the average of the background luminance (60.7 cd/m^2). For the red–green target we used the following seven levels of contrast: 2.1%, 4.0%, 7.6%, 14.5%, 27.6%, 52.5%, and 100%. For the blue–yellow target we used 6.2%, 9.9%, 15.7%, 24.9%, 39.6%, 62.9%, and 100%. During the experiment, the color target moves in a Brownian walk fashion within the background. For every frame during motion, each checker of the gray background is randomly refreshed with a random luminance around the gray point ($SD = 10\%$ contrast). The gray background is randomly jittered, instead of being uniform gray, because this design prevents from the participant from using any residual luminance artifacts to detect the color target (Hansen & Gegenfurtner, 2006; Stromeyer et al., 1999).

Procedure

Before the experiment, the participants were asked to read and sign an informed consent form. We ensured that the participants understood the experimental process and made it clear that they had the right to terminate the experiment whenever they wanted. The whole experiment was conducted in a dimly lit room.

Chromatic detection 2AFC task

For all participants, we first measured their detection threshold for red–green and blue–yellow stimuli with a temporal two-alternative forced-choice (2AFC) task. The red–green and blue–yellow stimuli were measured in two separate blocks, with 80 trials each. In each trial, there are two temporal intervals (1-s duration each) signaled by sound ("one" and "two" in Chinese). The color target appeared in one of the two intervals, and the participant had to press a key to indicate which interval contained the color target. The contrast of the color target was adaptively adjusted between

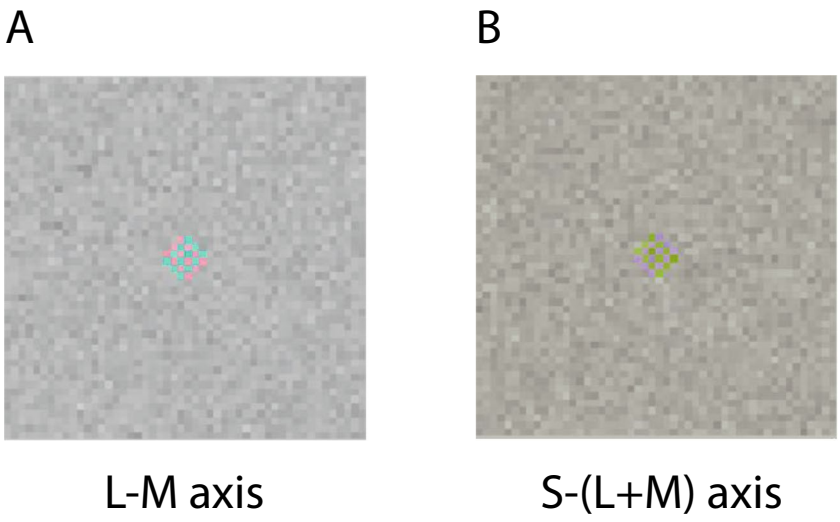

Fig. 2 Examples of color stimuli used in the present study. (**A**) The red–green (i.e., L–M axis) target at a contrast of 100%. (**B**) The blue–yellow target at a contrast of 100% (i.e., S–(L + M) axis). Both color targets have the same luminance as the average of the background (60.7 cd/m^2) on the monitor

trials using a QUEST procedure (Emerson, 1986; King-Smith et al., 1994).

Continuous motion tracking task

In the tracking task, the participant moved a digitizer on a trackpad to control a cursor on the screen, and continuously tracked a moving target by the cursor. The target made a Brownian walk, which was implemented with two Gaussian white noise sequences on the *x* and *y* axes, respectively. Each position was determined by the former position plus a random displacement which followed a normal distribution with a deviation of two pixels along each axis. At the beginning of each trial, a blue initial fixation point (0.1° × 0.1°) was presented in the center. When ready, the participant moved the cursor on the fixation point to start the trial. The color target (see Fig. 2) would appear at the center and start to move in a Brownian walk as depicted above. Each trial lasted 60 s. There were two colors for the target (red–green or blue–yellow) and seven levels of contrast, resulting in 14 conditions in total. Each condition had two trials, resulting in 28 tracking trials in total.

Data analyses

Analysis of the threshold in the chromatic detection 2AFC task

The responses of individual participants in the detection task were analyzed using the Quest toolbox, which is a part of Psychtoolbox (version 3.0.19). A threshold of 82% accuracy was fitted with the QuestMean function (King-Smith et al., 1994; Pelli et al., 1988) for each participant and each of the color conditions (i.e., red–green and blue–yellow) separately.

Analysis of the tracking data

We adapted Bonnen et al. (2015)'s method for the analysis of motion tracking data, with the exception of introducing two additional factors (i.e., latency from target motion to response and motor noise). A cross-correlation method (see Mulligan et al., 2013) was used to measure the latency from target motion to tracking response. To quantify individuals' sensory processing during tracking of the target, a Kalman filter model was used to estimate the observer's sensory noise, given all the target positions and the observer's tracking responses in each tracking trial. The posterior probability distribution of the parameter *Rs*, which refers to sensory processing noise variance, was estimated using Gibbs sampling and the JAGS package (Plummer, 2003), and its reciprocal, τs, was used as the measure of individuals' sensory sensitivity. Intuitively, *Rs* is an index that is comparable to tracking errors (i.e., deviations of response from presented target positions). However, the plausibility of using a Kalman filter to quantify tracking performance lies in the fact that the estimated sensory noise, rather than tracking errors, can be tightly correlated to sensory threshold measured by traditional psychophysics (Bonnen et al., 2015). For details of model construction and parameter estimation, please see in Appendix A.

Analysis of classification accuracy

To test whether tracking performance data can be used to discriminate CVD observers from observers with normal color vision, we performed a classification analysis using the receiver operating characteristic (ROC) curve (Fawcett, 2006). The area under the ROC curve (AUC), as well as an optimized classification criterion, was calculated to index the accuracy of the classification by estimated sensory sensitivity τs at different contrast levels.

Contrast-response function (CRF) analysis

To enable a more systematic comparison between sensory noise extracted from continuous motion tracking performance and color sensitivity measured through the traditional psychophysical task, we fitted the mean τs at different contrast levels to construct a CRF curve for each color axis condition, and performed the same fitting for each individual participant and each color axis condition. To be specific, we first rescaled each τs using the following equation:

$$\tau s'(c) = \frac{\tau s(c) - \tau s_{min}}{\tau s_{max} - \tau s_{min}} \quad (1)$$

where c refers to the stimulus contrast level, and τs_{min} and τs_{max} refer to the minimum and maximum reciprocals of estimated sensory noise for a certain participant and a certain color axis condition, respectively. Then, we fitted the rescaled reciprocals of estimated sensory noise, $\tau s'$, using a typical contrast response function which has been used in previous studies (Boynton et al., 1997; Tregillus et al., 2021):

$$\tau s'(c) = \frac{c^{p+q}}{c^q + c_{50}^q} \quad (2)$$

where p, q, and c_{50} are all unknown parameters to fit. Note that c_{50} refers to a contrast level with which τs would reach half of its maximum. We then tested the correlations between estimated threshold and fitted c_{50} to investigate these hypothetical relationships and further test the validity of the tracking task in measuring color sensitivity.

Results and discussion

Detection threshold

Figure 3A and B show the contrast thresholds of individual participants measured from the traditional 2AFC detection task at the L–M and S color axes. As expected, the CVD group had dramatically higher thresholds than those of the normal group. For the L–M axis, the mean contrast threshold of the CVD group (16.27% ± 4.00% *SD*) was 7.5 times that of the normal group (1.90% ± 0.97% *SD*), and an independent *t*-test showed a statistically significant difference between them ($t(27) = 13.51$, $p < 0.001$, Hedge's $g = 4.88$). For the S–(L + M) axis, the difference between mean thresholds was much smaller (CVD group: 9.35% ± 3.47% *SD*; normal group: 5.93% ± 1.75% *SD*), although the difference was statistically significant ($t(27) = 3.39$, $p = 0.002$, Hedge's $g = 1.22$).

Estimated sensory sensitivity from tracking

We asked CVD and control observers to track a moving target defined at the L–M or S color axis at various levels of contrast. A Kalman filter model was used to estimate the sensory sensitivity (τs) for each trial of the

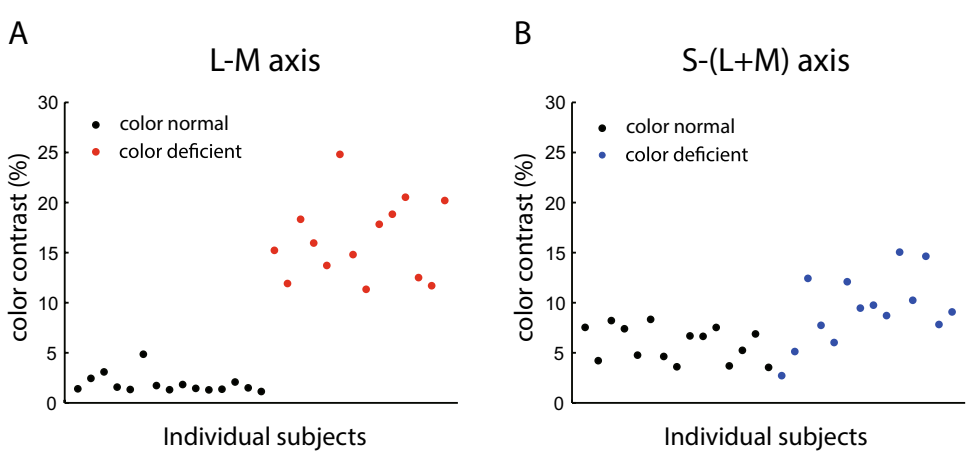

Fig. 3 The contrast thresholds of individual participants in the L–M axis (**A**) and S–(L + M) axis (**B**), respectively, measured from the traditional detection 2AFC task in Experiment 1

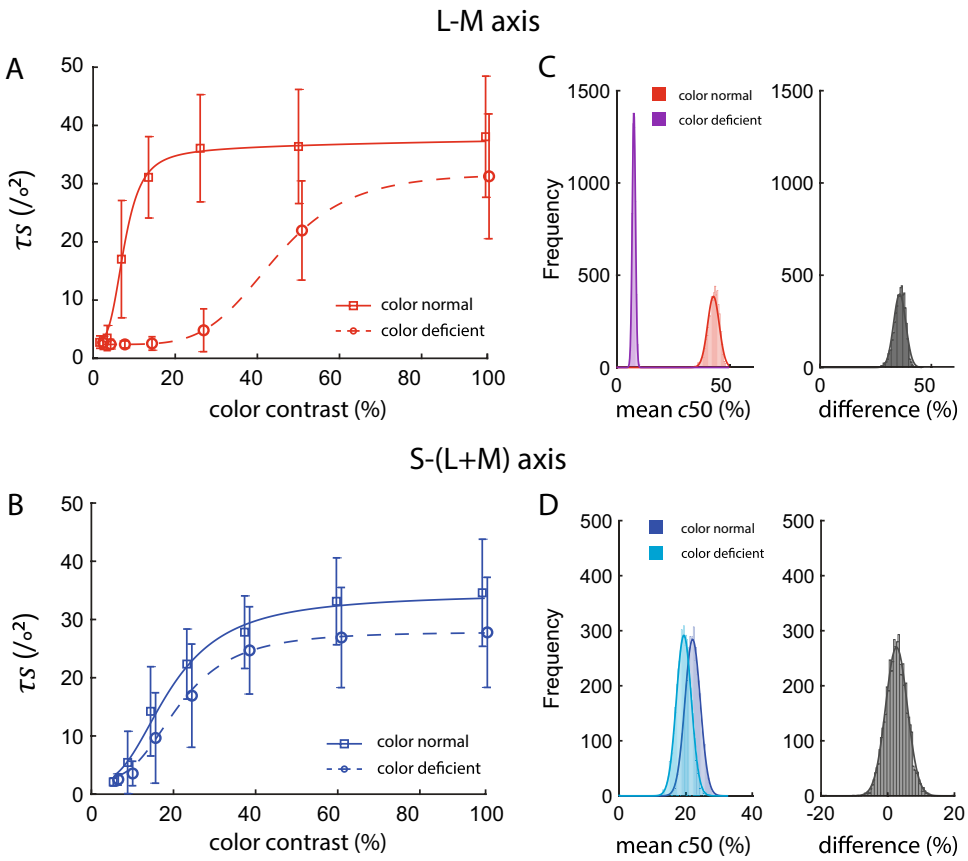

Fig. 4 Means of estimated τs as a function of color contrast levels in Experiment 1. **A** The upper-left panel plots estimated τs in the L–M axis (red), and **B** the lower-left panel plots estimated τs in the S–(L + M) axis (blue). The solid lines represent fitted functions of the normal individuals, and the dashed lines those of individuals with color vision deficiency. Error bars denote standard deviations across estimated τs of individual participants in each condition. **C** The upper-right panel shows the distribution of mean c_{50} for color-normal (red) and color-deficient (purple) individuals in the L–M axis, and the distribution of differences in c_{50} between the two groups. **D** The lower-right panel shows the distribution of mean c_{50} for color-normal (dark blue) and color-deficient (light blue) individuals in the S–(L + M) axis, and the distribution of differences in c_{50} between the two groups

tracking data. Figure 4A shows the mean τs for two groups as a function of contrast on the L–M axis. One can see that the two groups differ dramatically in τs, especially at the contrast levels of 8–28%. A mixed-design ANOVA revealed statistically significant main effects of contrast level ($F(6,162) = 143.94$, $p < 0.001$, $\eta_p^2 = 0.842$) and group ($F(1,27) = 67.67$, $p < 0.001$, $\eta_p^2 = 0.715$). In addition, the interaction between contrast level and group was statistically significant ($F(6,162) = 35.97$, $p < 0.001$, $\eta_p^2 = 0.571$). To further investigate how the group effects varied among contrast levels, seven t-tests were carried out separately as post hoc tests, and p values were adjusted by Bonferroni correction. The post hoc tests showed that the group effects were statistically significant at contrast levels of 8–53% ($t(27) \geq 4.21$, $p < 0.002$) but not at the lower contrast levels of 2% and 4% ($t(27) \leq 1.72$, $p \geq 0.686$), or the highest contrast of 100% ($t(27) = 1.73$, $p = 0.658$). The finding that tracking performance did not differ at low contrast of 2% and 4% was expected, since tracking a target at such low contrast levels would be almost impossible. The fact that the tracking performance for CVD and controls was comparable at high contrast (100%) is intriguing, and was probably due to compensatory mechanisms of color processing in CVD people (Boehm et al., 2014; Bosten et al., 2005; Macleod, 2003; Regan & Mollon, 1997; Tregillus et al., 2021; Webster et al., 2010). We further discuss this issue in the general discussion.

In the S–(L + M) axis conditions, the differences between the mean τs of two groups were much smaller (see Fig. 4B), although the main effect of group was still statistically significant ($F(1,27) = 4.34$, $p = 0.047$, $\eta_p^2 = 0.139$). The main effect of contrast was statistically significant ($F(6,162) = 164.17$, $p < 0.001$, $\eta_p^2 = 0.859$), as in the conditions of the L–M axis, but the interaction between contrast level and group was not statistically significant ($F(6,162) = 1.92$, $p = 0.080$, $\eta_p^2 = 0.066$).

We further performed AUC analysis to quantify how well tracking performance could distinguish CVD from

Table 2 Classification analysis for Experiment 1

		L–M axis				S–(L + M) axis			
		AUC	Optimized criterion	Hit rate	FA* rate	AUC	Optimized criterion	Hit rate	FA* rate
Detection task									
Contrast threshold		1.000	8.10%	100%	0%	0.829	7.64%	78.6%	13.3%
Tracking task									
τs (/deg^2)	Contrast Level 1	0.567	0.514	78.6%	46.7%	0.324	0.797	21.4%	13.3%
	Contrast Level 2	0.581	0.601	50%	33.3%	0.605	0.488	85.7%	46.7%
	Contrast Level 3	0.933	0.133	100%	13.3%	0.695	0.446	64.3%	26.7%
	Contrast Level 4	1.000	0.000	100%	0%	0.695	0.473	57.1%	20%
	Contrast Level 5	1.000	0.000	100%	0%	0.595	0.588	64.3%	46.7%
	Contrast Level 6	0.895	0.246	85.7%	20%	0.733	0.391	71.4%	26.7%
	Contrast Level 7	0.667	0.488	64.3%	33.3%	0.776	0.381	64.3%	13.3%

*FA—false alarm

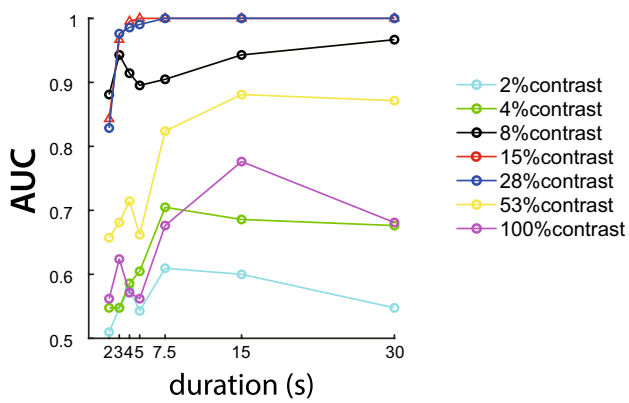

Fig. 5 The areas under the curves (i.e., AUCs) as a function of tracking duration at different color contrast levels in the L–M axis in Experiment 1

control observers. The AUC was higher than 0.93 at three contrast levels (8–28%) on the L–M axis (see Table 2), indicating that τs estimated with any of these color contrasts was a good index for distinguishing between people with CVD and those with normal color vision. The AUC at color contrasts of 15% and 28% reached 1. These results indicate that τs estimated with either of these color contrasts provided information with which the CVD observers could be perfectly distinguished from the observers with normal color vision, with zero misidentification. On the other hand, none of AUCs calculated by τs estimated with contrasts on the S–(L + M) axis surpassed 0.78.

The influence of tracking duration

The analysis above showed that tracking data with 15% and 28% red–green stimuli resulted in perfect classification (AUC = 1). Since the tracking duration in the experiment was relatively long, at 60 s, here we want to determine whether a shorter duration is sufficient for identifying color deficiency. We repeated the analysis using data with certain durations (from 2 to 60 s) for red–green stimuli. Figure 5 shows the result. Overall, AUC increased with longer duration. For both 15% and 28% conditions, a duration of 7.5 s or longer was sufficient to reach 100% accuracy (AUC = 1). Thus, this result suggests that 7.5 s is an optimal duration for future use of the current method for balancing accuracy and efficiency.

Contrast-response function analysis

We fitted a CRF for the mean τs as a function of color contrast along each color axis, as shown in Fig. 4A and B, and compared the fitted parameters from different groups and color axis conditions. To investigate whether these parameters were statistically different, we performed 5000 permutations, in each of which 14 and 15 participants were randomly selected from the color-normal and color-deficient groups with replacements, respectively, to form two random groups, and the parameters (i.e., p, q, and c_{50}) were then fitted and recorded. As mentioned above, we focused on what was most related to thresholds from traditional methods. For the L–M axis, fitted c_{50} had a 95% CI of [7.0%, 9.7%] for the color-normal group and [40.4%, 50.5%] for

the color-deficient group. The 95% CI of their difference was [31.9%, 42.6%], indicating the statistically significant difference in $c_{50\ \mathrm{along}}$ the L–M axis between the two groups (see also Fig. 4C). In comparison, the 95% CIs along the S–(L + M) axis were [15.1%, 23.8%] for the color-normal group and [17.7%, 26.4%] for the color-deficient group. The 95% CI of the difference between the two groups ([− 3.6%, 8.9%]) included 0, suggesting a lack of statistical significance (see also Fig. 4D). For the results of permutation analysis on p and q, please see Appendix B.

We then fitted CRFs for individual participants. Figure 6 shows fitted CRFs for two representative participants, one in the normal group (left) and the other in the CVD group (right), in two color axis conditions. Fitted CRFs for all the individual participants are presented in Fig. 15 in Appendix B. We compared the fitted parameters of c_{50}, as a measure of color contrast sensitivity from the tracking task, to the measured thresholds from the detection task for each individual participant and each color axis condition. Figure 7 plots the relationship between fitted parameters of c_{50} and measured detection thresholds of individual participants in two color axis conditions.

We then investigated the correlations between c_{50} and detection threshold. Considering the small samples in Experiment 1, we chose to compute skipped Pearson

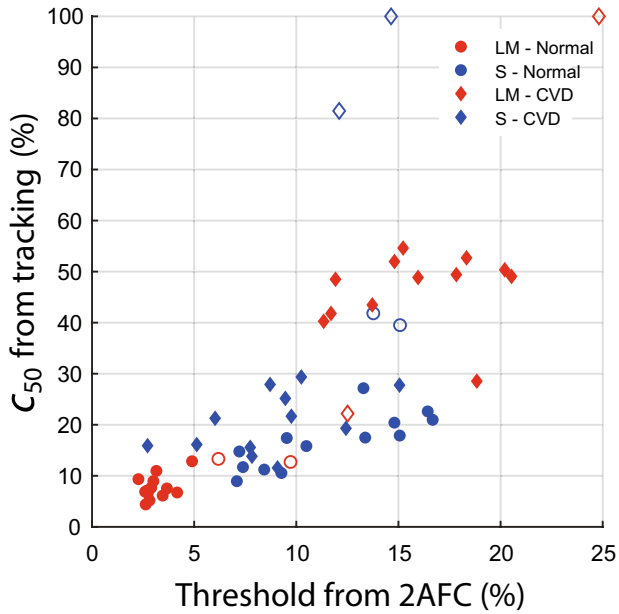

Fig. 7 Relationship between fitted c_{50} parameters and thresholds measured by the 2AFC task for individual participants (circles for color-normal, diamonds for color vision-deficient individuals) in two color axis conditions (red for the L–M axis; blue for the S–(L + M) axis) in Experiment 1. The solid and hollow plots indicate the normal values and the outliers identified by the boxplot rule, respectively

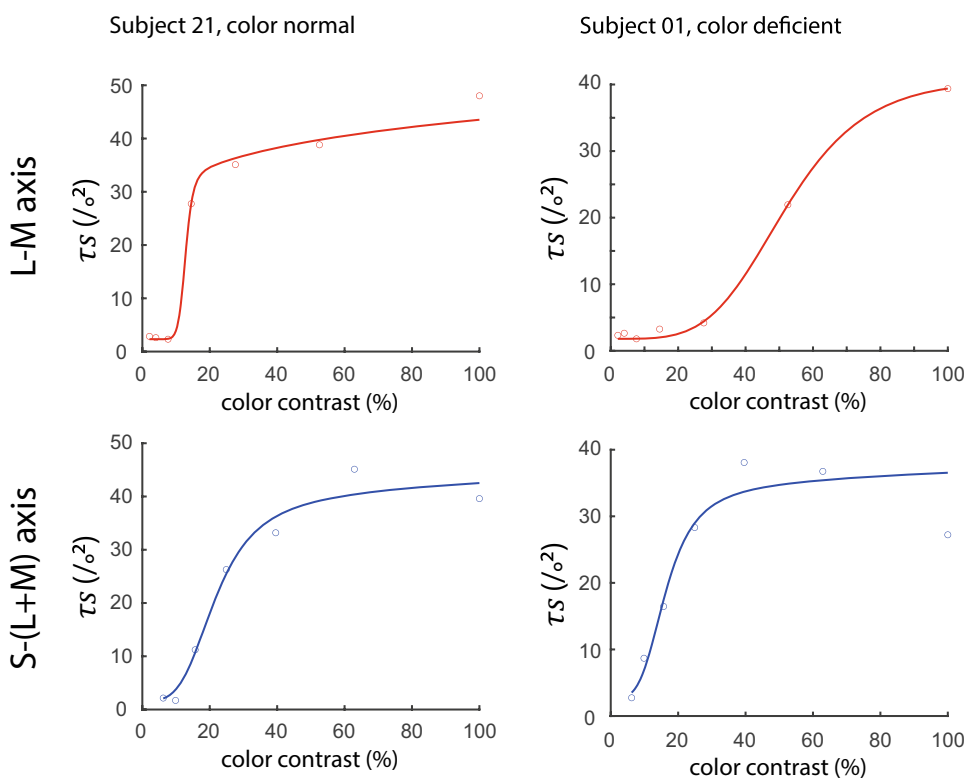

Fig. 6 Fitted contrast response functions (CRFs) based on estimated τs for two sample participants in Experiment 1. One is from the normal group (left) and the other is from the color vision-deficient group (right). The upper panels and lower panels show the fitted CRFs in the L–M axis and in the S–(L + M) axis, respectively

correlation coefficients (Pernet et al., 2012) rather than traditional Pearson correlation coefficients, to increase the robustness of the correlational analyses. We found moderate correlations between c_{50} and detection threshold for both groups and both color axis conditions (normal group, L–M axis: $r(13) = 0.473$; CVD group, L–M axis: $r(12) = 0.640$; normal group, S axis: $r(13) = 0.632$; CVD group, S axis: $r(12) = 0.585$). Note that the bootstrapped significance tests demonstrated that the correlations were statistically significant only in the S axis condition for the normal group and in the L–M axis condition for the CVD group, not in the other two conditions. The statistically nonsignificant correlation coefficients might be due to the outliers which were identified using the boxplot rule by Pernet et al. (2012) and are demonstrated by hollow plots in Fig. 7.

We also investigated whether the tracking performance of the CVD observers showed any compensatory enhancement in perception of colors along the L–M axis, as demonstrated by the neural imaging results from Tregillus et al. (2021). To this end, we adapted the method for constructing a reduction model by Tregillus et al. Specifically, we first fitted the means of τs at the different contrast levels of the color-normal observers using Eq. 2 to construct CRFs for the whole group of color-normal observers. We then rescaled the CRFs according to the ratio between the mean detection thresholds of the two groups (i.e., constructing the reduction models) and used the reduction models to predict the mean τs of the CVD observers at the highest contrast levels along different color axes.

Figure 8 shows how the reduction models predict the mean τs of the CVD group at different color contrast levels. For the L–M axis, we found that the mean τs predicted by the reduction model was 23.03, beyond the 95% CI of observed mean τs (i.e., [25.07, 37.44]), indicating that the CVD group had a compensatory accelerated increase when color contrasts were suprathreshold. To the contrast, the predicted τs (i.e., 26.09) was inside the 95% CI of the observed mean τs (i.e., [22.33, 33.24]), at the highest contrast along the S axis.

Experiment 2

The results of Experiment 1 showed substantial differences in tracking performance between the CVD group and color-normal groups, allowing for a perfect classification between the two groups. In Experiment 2, we aimed to further examine the effectiveness and efficiency of the new screening method for identifying color vision-deficient people, using a larger sample ($N = 171$) and a shorter duration (20 s). Another notable difference between the two experiments is that participants for Experiment 2 were randomly recruited from the general population without pre-screening. Our objective was to determine whether the new method could successfully identify the CVD in a more realistic situation. Additionally, we reduced the number of contrast conditions to four levels for each color axis and included only one trial for each condition in order to streamline the overall procedure and concentrate on the more discriminative contrast levels, i.e., 15% and 28% on the L–M axis.

Method

Participants

A total of 171 participants (109 male, 62 female, average age: 21.6 years, age range: 18–31 years) were recruited for the experiment. All participants were recruited at a

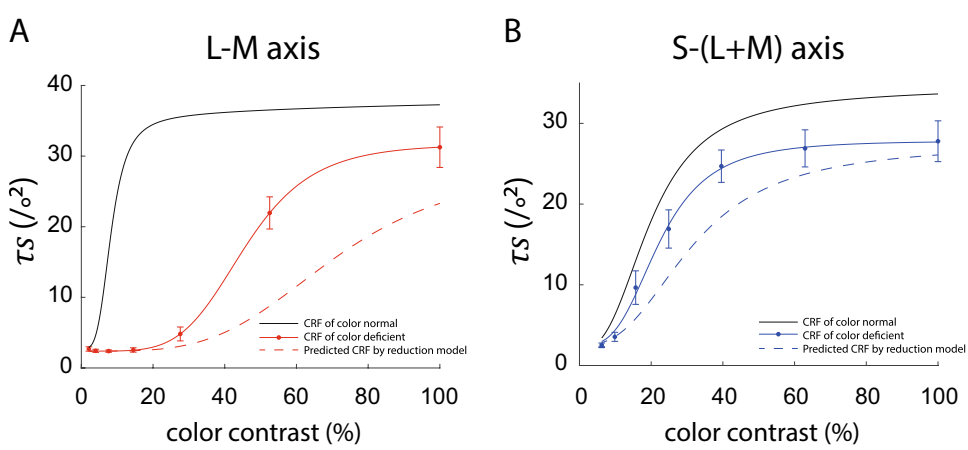

Fig. 8 Comparison between the contrast response functions (CRF) predicted by the reduction model (colored dashed line) and the fitted CRFs of the color-deficient observers (colored solid line). The CRFs of the color-normal observers are also presented as reference (black solid line). Error bars denote standard errors of the mean τs of the color-deficient observers

university in Shanghai, PR China. Participants were not screened by the Ishihara test before the experiment, but received the test after the experiment. Among the group, 12 participants (all male) were identified as having CVD. Eight of the CVD observers were tested by an anomaloscope, and the results showed that one exhibited protanomaly, four deuteranomaly, and three protanopia. Participants were compensated for their participation. They signed written informed consent forms in accordance with the Declaration of Helsinki. The experimental procedure was approved by the ethics committee of Shanghai University of Sport (No. 102772019RT009).

Apparatus and stimuli

The apparatus and the stimuli were the same as those used in Experiment 1. One exception was that the number of contrast levels used in Experiment 2 was reduced to four for each color axis (L–M axis: 7.6%, 14.5%, 27.6%, 52.5%; S–(L + M) axis: 15.7%, 24.9%, 39.6%, 62.9%).

Procedure

Participants in Experiment 2 only performed the tracking task, which followed the same procedure as the tracking task in Experiment 1, with the exception that each trial in Experiment 2 lasted for 20 s.

Results and discussion

Estimated sensory sensitivity from tracking

As in Experiment 1, we used the same Kalman filter model to estimate sensory sensitivity (τs) for each trial. Based on the results from the Ishihara test, 12 of 171 participants were classified as CVD (also see the Method section). Figure 9 shows the mean τs of these CVD participants in different conditions of color axes and contrast levels, compared with the mean τs of the remaining 159 color-normal participants. A mixed-design ANOVA showed that, for the L–M axis, both the main effects of contrast level and group (CVD vs. color-normal) were statistically significant (contrast level: $F(3,507) = 104.839$, $p < 0.001$, $\eta_p^2 = 0.383$; group: $F(1,169) = 78.587$, $p < 0.001$, $\eta_p^2 = 0.317$). The interaction between group and contrast level was statistically significant ($F(3,507) = 15.415$, $p < 0.001$, $\eta_p^2 = 0.084$). A series of post hoc tests also showed a statistically significant difference in τs in each contrast level between the two groups ($t(169) \geq 2.904$, $p < 0.004$). In contrast, there was a statistically significant main effect of contrast level only in the conditions of S–(L + M) axis ($F(3,507) = 66.134$, $p < 0.001$, $\eta_p^2 = 0.281$). The main effect of group was not statistically significant ($F(1,169) = 3.290$, $p = 0.071$, $\eta_p^2 = 0.019$), and the interaction between contrast level and group was not statistically significant ($F(3,507) = 1.696$, $p = 0.167$, $\eta_p^2 = 0.010$). All these results were consistent with the findings in Experiment 1, indicating differences in color sensitivity between the two groups in the L–M axis conditions but not the (L + M)–S axis conditions.

We then analyzed the AUCs of τs for distinguishing between CVD and color-normal groups in the L–M axis conditions. As shown in Fig. 10, the AUCs were 0.807, 1, 0.896, and 0.806 at contrast levels of 7.6%, 14.5%, 27.6%, and 52.5%, respectively (for details, see Table 3). As in Experiment 1, the AUC reached 1 at the contrast level of 14.5%, indicating excellent classification of CVD with the participants' tracking performance at this contrast level. To conduct a more comprehensive analysis of the distribution of τs among individual participants, we plotted and compared the histograms of τs for the contrast level condition of 14.5%

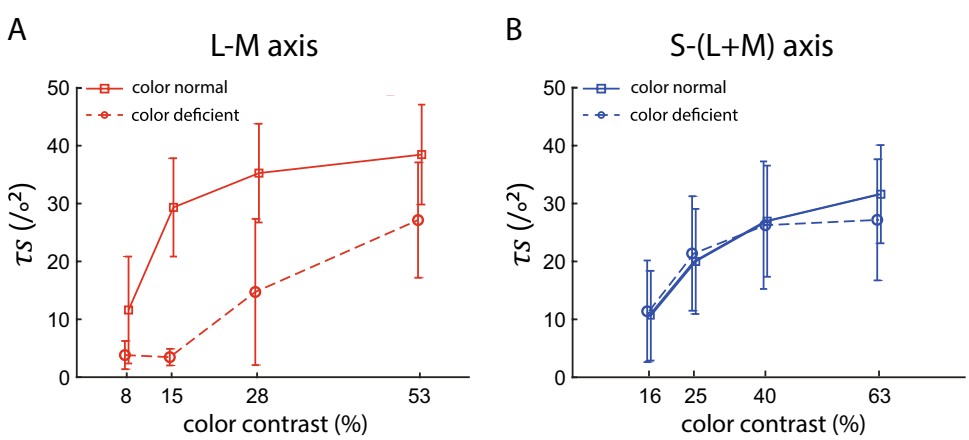

Fig. 9 Means of estimated τs as a function of color contrast levels in Experiment 2. **A** The left panel plots estimated τs in the L–M axis, and **B** the right panel plots estimated τs in the S–(L + M) axis. The solid lines represent the color-normal group and the dashed lines the color vision-deficient group. Error bars denote standard deviations across estimated τs of individual participants in each condition

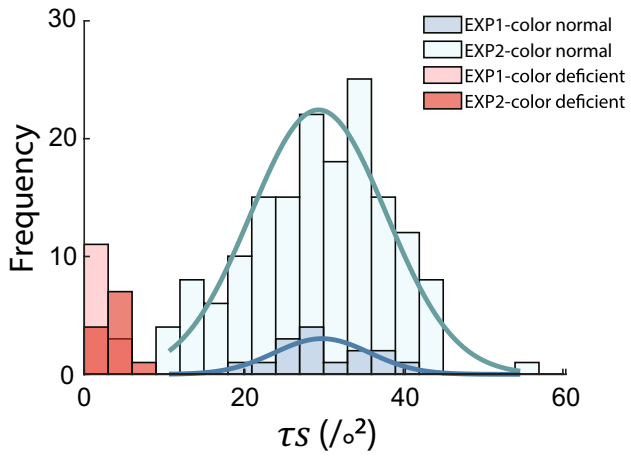

Fig. 10 The areas under the curve (AUCs) as a function of color contrast level in the L–M axis in Experiment 2

in the L–M axis, where the color-normal and CVD groups showed the largest differences in τs. As Fig. 11 shows, the τs of the color-normal participants in both Experiments 1 and 2 had close-to-Gaussian distributions and similar ranges (95% CI: 22.4–37.0 for Experiment 1; 22.2–30.9 for Experiment 2). In contrast, the τs of the CVD participants from both experiments were all close to zero, indicating that they were insensitive to color information in such conditions, as the Ishihara test demonstrated.

In brief, the results from Experiment 2 were highly consistent with the findings in Experiment 1, both of which indicated that estimated sensory sensitivity from motion tracking provides sufficient information for screening CVD, comparable to that of the typical clinical tests.

Test–retest reliability

To test the reliability of the assessment by motion tracking, we reinvited 18 participants who had participated in either Experiment 1 ($N=7$, all color-normal observers) or Experiment 2 ($N=11$, all color-normal observers), 3 months after they had finished their first experiments. They performed the same task as in Experiment 2. We compared the estimated τs from this "retest" experiment with those from the previous experiments. For the participants from Experiment 1, τs was re-estimated using the data for the first 20 s of each trial and averaged between two trials for each condition so as to match the conditions between the experiments.

Figure 12 shows the results from the test and the retest. The mean τs appeared to be highly consistent in each of the conditions between the two tests. Two repeated-measures ANOVAs showed that neither the main effects of test–retest nor the interactions between test–retest and contrast levels were statistically significant for either color axis [test–retest, L–M axis: $F(1,17)=0.942$, $p=0.339$, $\eta_p^2=0.027$; test–retest × contrast, L–M axis: $F(1,17)=0.249$, $p=0.839$, $\eta_p^2=0.007$; test–retest, S–(L + M) axis: $F(1,17)=0.554$, $p=0.462$, $\eta_p^2=0.016$; test–retest × contrast, S–(L + M) axis: $F(1,17)=0.658$, $p=0.560$, $\eta_p^2=0.019$], while the main effect of contrast was statistically significant (L–M axis: $F(3,51)=124.724$, $p<0.001$, $\eta_p^2=0.786$; S–(L + M) axis: $F(3,51)=110.900$, $p<0.001$, $\eta_p^2=0.765$). The correlational analysis also showed moderate-to-strong correlations between the estimated τs from

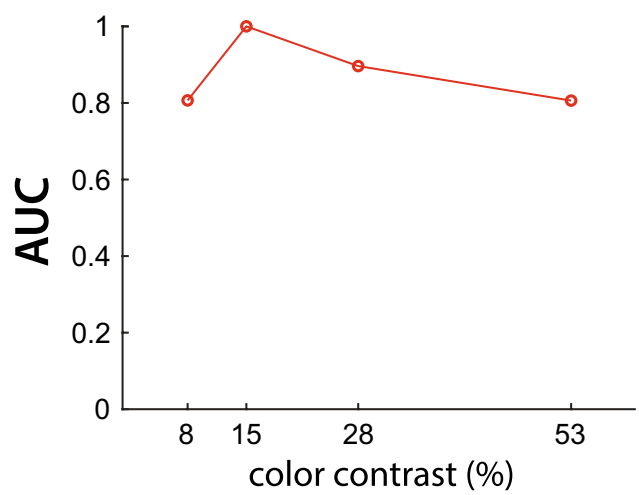

Fig. 11 Distribution of estimated τs for individual participants in Experiments 1 and 2 with fitted Gaussian curves for color-normal observers in Experiments 1 and 2, respectively

Table 3 Classification analysis for Experiment 2

		L–M axis				S–(L + M) axis			
		AUC	Optimized criterion	Hit rate	FA* rate	AUC	Optimized criterion	Hit rate	FA* rate
τs (/deg²)	Contrast Level 1	0.807	4.2574	77.4%	16.7%	0.486	10.9948	39.6%	33.3%
	Contrast Level 2	1.000	8.3195	100%	0%	0.464	23.6058	34.6%	25%
	Contrast Level 3	0.896	24.3207	91.8%	25%	0.498	19.3485	77.9%	58.3%
	Contrast Level 4	0.806	33.5688	79.2%	25%	0.613	29.5052	66.0%	41.7%

*FA—false alarm

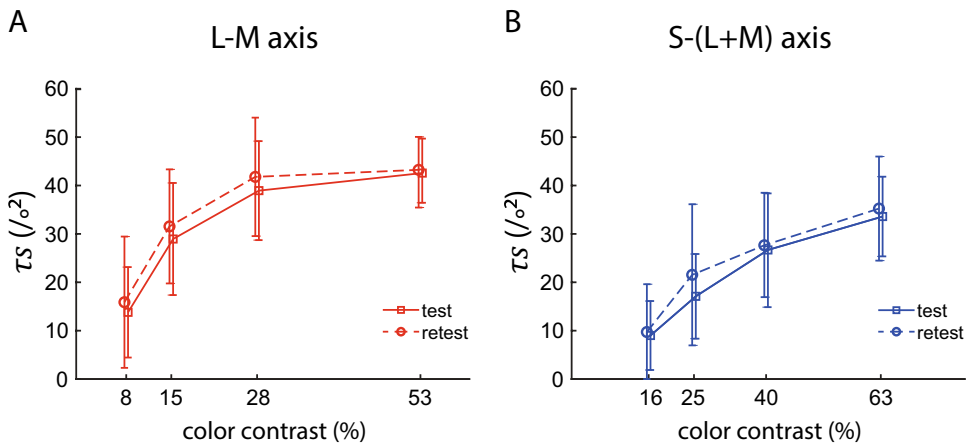

Fig. 12 Means of estimated τs as a function of color contrast levels from the test and the retest. **A** The left panel plots estimated τs in the L–M axis (red) and **B** the right panel plots estimated τs in the S–(L + M) axis (blue). Error bars denote standard deviations across estimated τs of individual participants in each condition

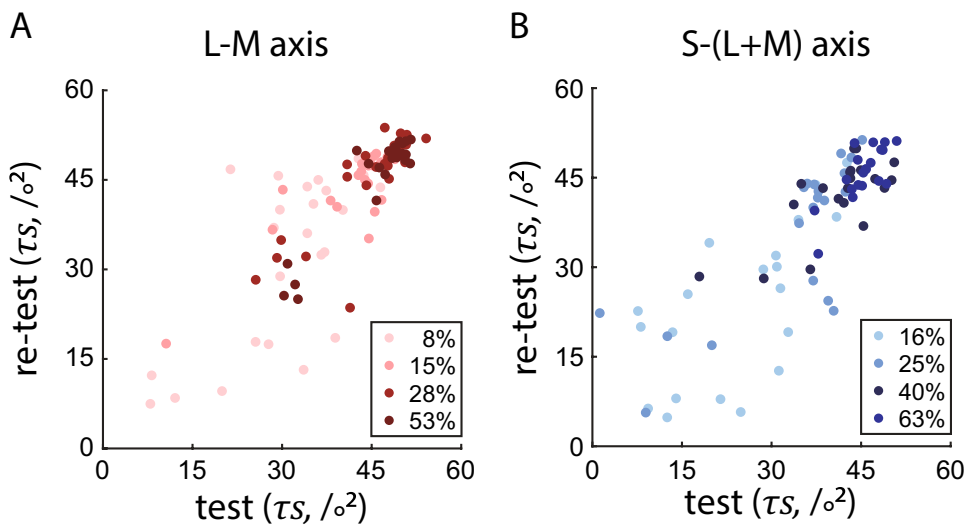

Fig. 13 The graph shows the relationship between τs from the test and the retest for each participant of each contrast on the L–M axis (**A**, reddish plots) and the S–(L + M) axis (**B**, blueish plots)

the two experiments across all eight conditions of different color axes and color contrasts ($rs(16)$: 0.611–0.852, $p < 0.007$; see also Fig. 13). Both sets of results indicate that the motion tracking method is reliable in estimating sensory sensitivity to color information.

General discussion

Motion tracking: A reliable and efficient tool for screening color vision deficiency

An important finding in the current study was that the performance of motion tracking between the CVD group and the normal group was dramatically different when color contrasts were defined on the L–M axis, whereas their performance was comparable with color contrasts on the S–(L + M) axis (see Fig. 4). These results were highly consistent with the screening results by the Ishihara test, according to which the CVD group should be deficient in discriminating red–green stimuli but not in discriminating yellow–blue stimuli. The follow-up AUC analyses further demonstrated that participants' performance in the motion tracking task provided sufficient information to screen CVD. As Figs. 4 and 5 show, perfect screening occurred on individuals' tracking performance toward 14.5% red–green stimuli, as indexed by an AUC of 1, in

both experiments. All these results indicated the effectiveness of motion tracking as a tool for CVD screening.

Besides its effectiveness, motion tracking also showed its efficiency. Despite the fact that each trial of the task was designed to last 1 min in Experiment 1, the analysis revealed that collecting individuals' motion performance at a contrast level of 14.5% or 27.6% for merely 7.5 s was sufficient to achieve a perfect classification of those with CVD. This finding was further confirmed by Experiment 2, in which each trial lasted 20 s and perfect screening was achieved at the contrast level of 14.5%. Considering its effectiveness and efficiency together, our results clearly indicate that the motion tracking task used in the current study has potential as an assessment tool in clinical screening comparable to the pseudoisochromatic plate tasks.

As an alternative method for CVD screening, assessment based on motion tracking has two unique advantages over the pseudoisochromatic plate tests. First, compared to most of the pseudoisochromatic plate tests, which are pre-made, stimuli used in this task are all generated randomly online and apparently prevent examinees from cheating by preparing for the test in advance. Another advantage is that this task is objective, and no verbal report is needed. For the pseudoisochromatic plate tasks, examinees need to recognize and report objects (e.g., numbers, shapes, or animals) depicted by color contrasts (Jurasevska et al., 2014; Rodriguez-Carmona et al., 2012), which could be problematic for some special groups with language deficits or intellectual disability. In contrast, motion tracking is an intuitive task and more feasible for these groups. Regarding these advantages, we think that motion tracking holds great potential as a tool for CVD screening and is worth further exploring and developing.

Our results revealed some unexpected difference between color-normal and color-deficient observers along the S axis. The detection thresholds along the S axis showed significant, albeit very small, differences. The tracking performance along the S axis in Experiment 1 tended to be different between color-deficient and color-normal observers as well. One possible reason could be any imprecision in color control due to adaptation, calibration, or display issues, especially considering that precise calibration of LED screens of laptops is usually difficult. We think that this is most unlikely, since the same device and setup was used in both experiments, but only the result of Experiment 1 showed a significant effect along the S axis. A second possible reason is that color-deficient observers have a slightly different S axis than color-normal individuals. For anomalous trichromats, they have an altered cone photopigment L′ or M′; their S axis is not S–(L + M), but instead S–(L′ + M) or S–(L + M′) (Bosten, 2019; Jordan et al., 2010). The difference is miniscule, but in theory it could explain the significantly different thresholds along the S axis in Experiment 1. Finally, individual differences among observers may also contribute to this issue. There are variations in isoluminance among individuals. We did not determine isoluminance for each observer but used luminance noise to mask residual luminance artifacts (Hansen & Gegenfurtner, 2006; Stromeyer et al., 1999). Additionally, individual differences in attention, motivation, and fatigue could be factors contributing to the unexpected group effect along the S axis. However, since the primary aim of this study was to apply the tracking task in practical and clinical settings, it is not feasible to control for all these individual differences in practice. Thus, the current results demonstrate that our method can tolerate these potential individual differences. Despite these issues, we believe our new method is robust enough to achieve perfect identification of CVD with zero misidentification.

Using motion tracking to study color vision in general

Another aim of the current study was to examine whether motion tracking could be used to comprehensively measure individuals' color vision. To compare the motion tracking task against the traditional psychophysical method, we fitted the participants' tracking performance using CRFs and estimated the parameter, c_{50}. The correlational analyses showed that c_{50} and the psychophysical threshold were moderately correlated ($r = 0.473$–0.640) within each homogeneous group (i.e., color-normal and CVD groups) and within each color condition (i.e., L–M and S–(L + M)), indicating a potential close relationship between the two methods in measuring color contrast sensitivity, which nevertheless needs to be confirmed with larger samples.

The close relationship between the performance in the dynamic motion tracking and the traditional psychophysical discrimination tasks is consistent with previous findings. For example, multiple studies (Bonnen et al., 2015; Dorr et al., 2017; Mooney et al., 2018) have reported moderate-to-strong correlations between luminance contrast sensitivity measured by traditional psychophysical methods and perceptual noise measured by dynamic motion tracking tasks. This line of evidence strongly indicates that dynamic motion tracking is an effective method to reveal individuals' fundamental visual functions such as processing of luminance contrast information. The findings from the present study have extended the boundary of this method by demonstrating that the dynamic motion tracking task can also be used to measure individuals' sensitivity to chromatic contrasts, indicating that motion tracking can serve as a general method for measuring visual processes, rather than one specific to some particular visual features. Such generality of motion tracking tasks is further confirmed by a recent study (Ambrosi et al., 2021) reporting the effectiveness of a motion tracking task in measuring individuals' numerosity representation. All these findings suggest that dynamic motion tracking has the potential to become a general method for measuring diverse visual processes, as traditional psychophysical methods are.

Of course, there are still nonnegligible differences in the chromatic contrast sensitivity estimated from motion tracking and traditional psychophysical methods. In particular, the estimated c_{50} is substantially larger than the threshold, and their correlation is only moderate, not strong (i.e., > 0.8). This could be due to multiple factors. First, as the two-stream hypothesis (Goodale & Milner, 1994; Milner & Goodale, 2008) has proposed, visual processing for perception and visuomotor control may be driven by different underlying neural mechanisms. There is ample empirical evidence both for (Goodale & Milner, 1994; Graziano & Gross, 1998; James et al., 2003; Rizzolatti & Luppino, 2001; Rizzolatti & Matelli, 2003; Rossetti et al., 2003) and against (Andersen, 1987; Franz et al., 2000; Ganel & Goodale, 2003) this hypothesis. Here, we have no intention of becoming involved in the controversy but raise the possibility that visuomotor tasks, such as dynamic motion tracking, may partially rely on different visual processes compared to traditional detection or discrimination tasks. Although performance in two types of tasks can reflect some shared information (e.g., luminance contrasts and chromatic contrasts), one may not expect perfectly congruent performance to be observed.

Second, response errors could potentially reduce the correlation between the two measurements. In contrast to the traditional tasks, which require discrete responses, dynamic motion tracking requires participants to make continuous movements, or motor judgements, in response to ongoing target displacement. Because of motor noise, more response errors can occur in motion tracking tasks (Bhat et al., 2018; Bonnen et al., 2015). This possibility is also supported by the present result wherein the estimated sensory noise (i.e., reciprocal of root-squared sensory sensitivity) by the motion tracking task did not approximate to 0 but rather a constant positive value with the increase in chromatic contrast, even for the normal color vision group. Moreover, strategies and subjective beliefs regarding visuomotor control might also introduce extra response bias, thus increasing response errors. To resolve these issues, Straub and Rothkopf (2022) reviewed and reanalyzed some previously published data using their modified models. For the new models, internal behavioral costs and subjective beliefs were included in considerations, and cost functions related to these factors were added into the structure of the model. These processes successfully minimized the differences between sensitivity estimation using traditional tasks and visuomotor tasks. However, we think that the use of cost functions in visuomotor analysis requires further empirical justification because the assumptions of internal behavioral costs and subjective beliefs are not guaranteed. Therefore, we chose not to use their model for the data analysis in this study.

Finally, the complicated relationship between motion and color must be considered as well. Motion perception and motor responses are impaired at isoluminance when the target and background do not contain luminance contrast (Cavanagh, 2011; Seiffert & Cavanagh, 1999). Thus, compared to the psychophysical threshold, the motion tracking task would in principle underestimate chromatic contrast sensitivity to a certain degree.

Despite the potential disagreements between dynamic motion tracking and traditional psychophysical methods in measuring chromatic contrast sensitivity, a significant advantage of motion tracking tasks is their much higher efficiency. Both previous studies (Bonnen et al., 2015; Mooney et al., 2018) and the current study show that a duration of 15–20 s is sufficient for the motion tracking task to provide a reliable estimation of perceptual noise at a certain color contrast. This advantage allowed us to construct CRFs for color vision of an acceptable duration of time (e.g., 2–2.5 min for estimating perceptual noise at 6–7 color contrast levels). In contrast, one might need at least 80–100 trials and over 5 min for a reliable estimation of threshold when using adaptive psychophysical methods. Note that adaptive procedures merely provide thresholds and responses to stimuli close to thresholds, while CRFs provide more comprehensive depiction of color perception, especially in the suprathreshold range. The methods using constant stimuli might be more comparable to the manual tracking task, which nevertheless requires more trials (e.g., 200–300 trials) and consumes more time (e.g., 10–15 min). In light of these factors, we consider that the manual tracking method is clearly more efficient than traditional psychophysical methods in depicting an individual's color vision.

Suprathreshold color vision assessed by motion tracking

Unlike the threshold-detection approach of traditional psychophysical methods (King-Smith et al., 1994; Kontsevich & Tyler, 1999; Pelli & Bex, 2013; Taylor & Creelman, 1967; Watson & Pelli, 1983; Watt & Andrews, 1981; Wetherill & Levitt, 1965), the motion tracking task is able to obtain observers' responses both when the contrasts of stimuli are close to threshold and when they are at suprathreshold levels. This makes it a useful tool to investigate the potential differences in color processing at threshold and suprathreshold levels. It has been proposed that the processing of threshold and suprathreshold colors might rely on distinct underlying mechanisms (Chickering, 1967; Kuehni, 1999, 2000).

Interestingly, our current study indeed revealed such a difference between threshold and suprathreshold vision. For stimuli with color contrast near threshold, CVD observers' tracking performance was much worse than that of controls. However, when color contrasts were higher than their threshold, their performance improved more rapidly with the color contrast increase than the prediction of the reduction model which was constructed based on the CRF of the color-normal people and rescaled by the ratio between the detection thresholds of the two groups (Tregillus et al., 2021; see also Fig. 8). This result indicates the

compensatory perception enhancement toward the suprathreshold color for people with CVD, and is consistent with recent findings that CVD observers' visual processing of suprathreshold color is not as impaired at their processing at threshold levels (Baraas et al., 2010; Boehm et al., 2014; Bosten et al., 2005; Foster & Nascimento, 2023; MacLeod, 2003; Regan & Mollon, 1997; Tregillus et al., 2021; Webster et al., 2010). Here, our findings showed that such compensation can also be revealed in the visual motor domain. The consistency between the previous and the current findings further confirms the phenomena of compensation of color processing and validates the efficacy of our method for assessing suprathreshold color processing.

Appendix 1

This appendix describes the Kalman filter model and the corresponding assumptions which were used to fit the sensory noise of individual participants when tracking patch with different color contrast levels in the continuous motion tracking task. The general concept of the analysis was presented in the subsection *Analysis of the tracking data*.

For each trial, we assume that an individual participant's response at a certain moment t, r_t follows a normal distribution:

$$r_t \sim N\left(\hat{r}_t, R\sum_t\right) \tag{3}$$

$$R\sum_t = Re_t + Rm \tag{4}$$

in which $\hat{r}_t$ refers to the estimated position at the moment t and $R\Sigma_t$ refers to the conjunct noise variance from position estimation at the moment t, Re_t, and motor execution, Rm. According to the Kalman filter model, $\hat{r}_t$ and Re_t are both determined with the Kalman gain K_t:

$$\hat{r}_t, = K_t s_{t-\Delta t} + \left(1 - K_t\right) r_{t-1} \tag{5}$$

$$Re_t = K_t Rs \tag{6}$$

in which $s_{t-\Delta t}$ refers to the presented target position (i.e., sensory input) at the moment t–Δt and Δt refers to latency from target motion to tracking. Δt for each trial was computed separately using the cross-correlation method. Rs refers to the noise variance from sensory processing and all the tracking data of this trial. K_t was then determined through a standard Kalman filter recurrent process:

$$K_t = \frac{RE_{t-1} + Q}{RE_{t-1} + Q + Rs} \tag{7}$$

in which Q refers to target displacement variance between frames. Regarding the experimental design wherein the target made a Brownian walk with a deviation of two pixels between frames (see also the subsection *Continuous motion tracking task*), Q is hypothetically set as 4 ($=2^2$) for posterior sample generation.

For model construction, note the following two points. First, we arbitrarily set t_0 as the moment just before the tracking began and set the hypothetical response at t_0, $r_{0,}$ as the coordinates of the screen center. Second, regarding the latency from target motion to tracking, we assume that participants' performance at the beginning of each trial (i.e., $t < \Delta t$) was not influenced by sensory input but merely determined by the response at the last moment and noise from motor execution:

$$r_t \sim N\left(r_{t-1}, Rm\right), when\ t < \Delta t \tag{8}$$

The posterior probability distributions of the unknown parameters (Rs, Rm, and K_t) were estimated from the data using Gibbs sampling to generate a Markov chain Monte Carlo (MCMC) chain of posterior samples. To estimate posterior distributions, a dispersive gamma distribution with $\alpha = 0.001$ and $\beta = 0.001$ was used as an uninformative prior distribution for Rs and Rm, respectively. For each sequence of responses on either the x or y axis of each trial, an MCMC chain of 5000 posterior samples was generated using the JAGS package (Plummer, 2003). The means of these samples were computed as the estimation of the unknown parameters. We then averaged parameters on the x and y axes for further analysis.

Appendix 2

The permutation analysis demonstrated that p fitted from the mean τs had a 95% CI of [0.006, 0.072] for the color-normal group and a 95% CI of [−0.077, −0.015] for the color-deficient group along the L–M axis. The 95% CI of their difference was [−0.127, −0.032]. Along the S–(L + M) axis, the 95% CIs of p were [−0.027, 0.089] for the color-normal group and [−0.039, 0.085] for the color-deficient group. Their difference had a 95% CI of [−0.099, 0.075]. The fitted q had a 95% CI of [3.205, 5.909] for the color-normal group and a 95% CI of [3.947, 6.040] for the color-deficient group along the L–M axis. The 95% CI of their difference was [−1.394, 2.261]. Along the S–(L + M) axis, the 95% CIs of q were [2.254, 3.490] for the color-normal group and [2.757, 4.158] for the color-deficient group. Their difference had a 95% CI of [−0.389, 1.569]. Figure 14 also plots the distributions of fitted p, q and their differences between the color-normal and color-deficient groups along different color axes by permutation analysis.

Figure 15 plots the estimated τs at different color contrast levels and the fitted contrast-response functions (CRFs) based on estimated τs for all the individual participants.

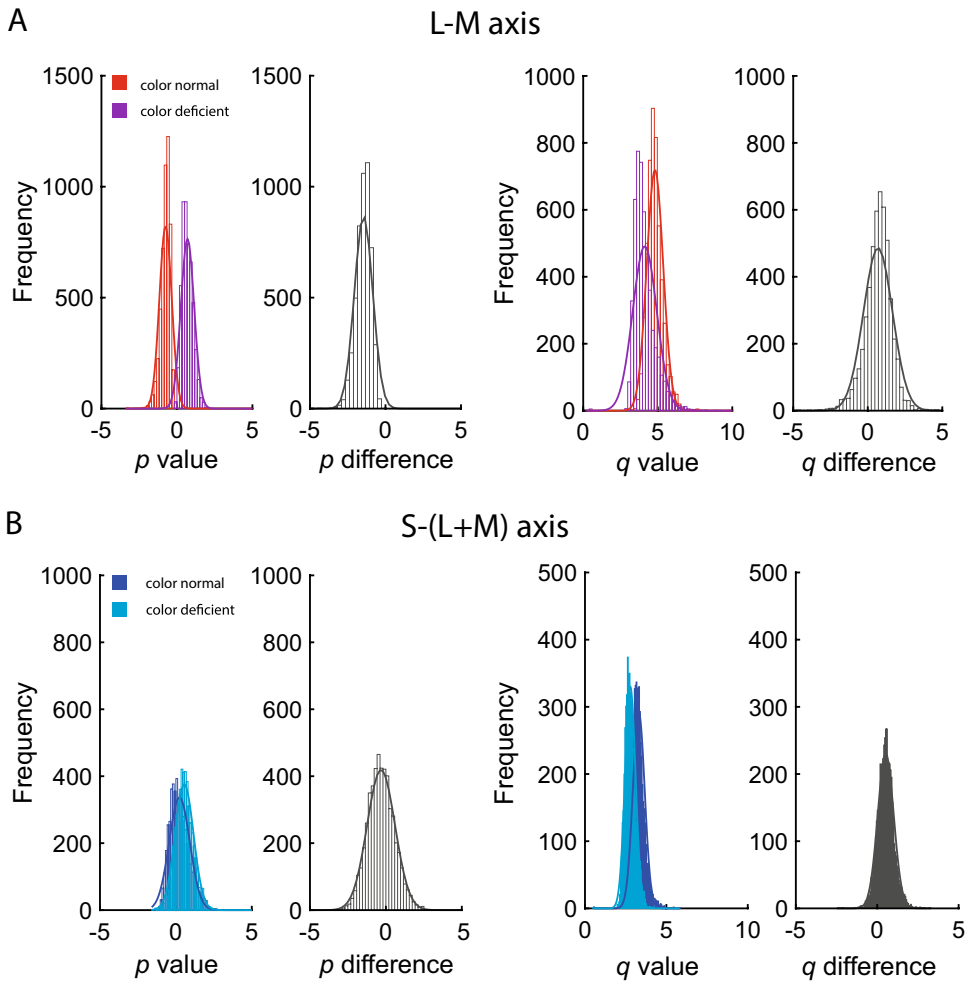

Fig. 14 Distributions of fitted p, q, and their differences between color-normal and color-deficient groups across the L–M axis (red: upper panel **A**) and S–(L + M) axis (blue: lower panel **B**)

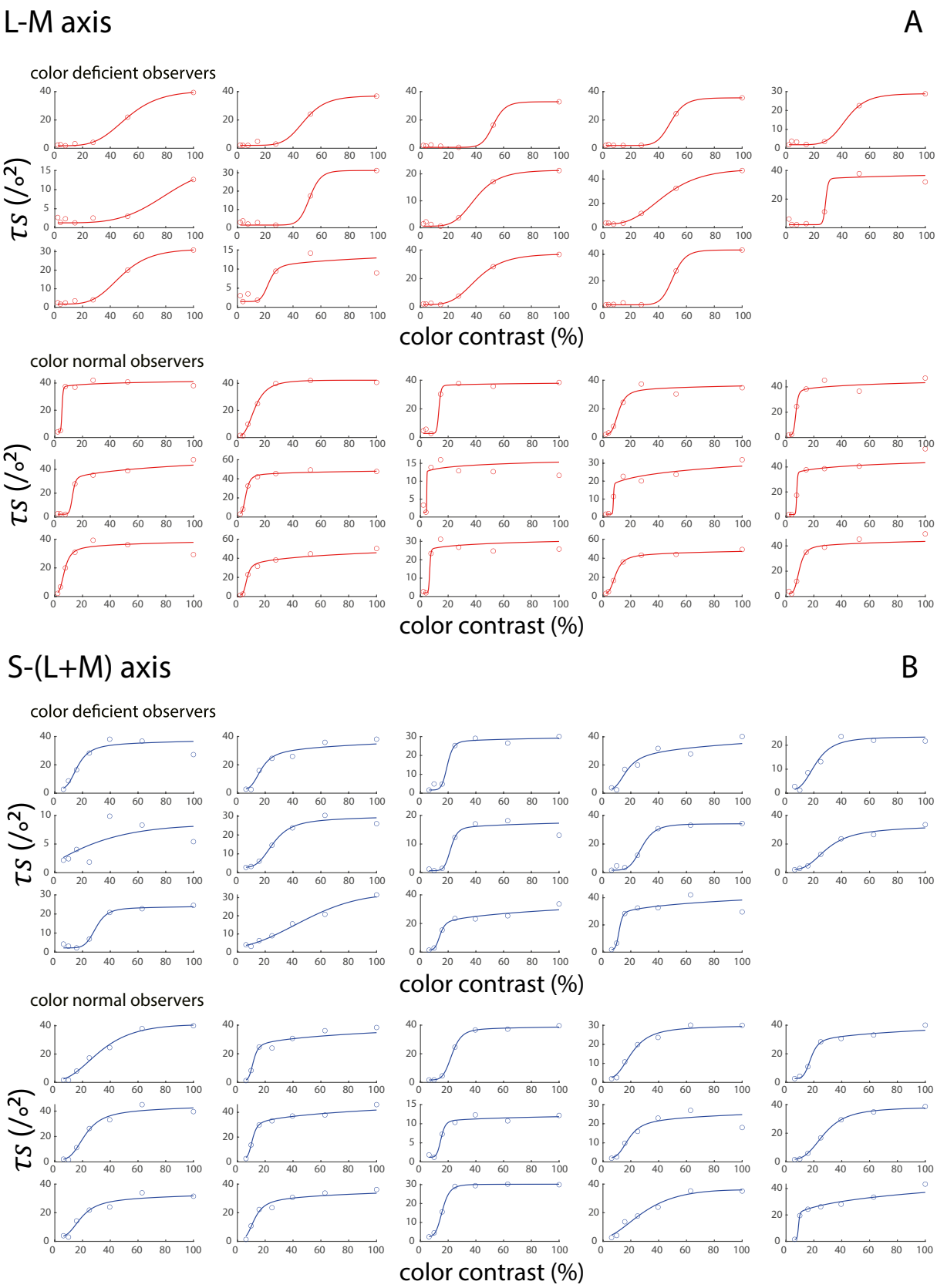

Fig. 15 The fitted contrast-response functions (CRFs) based on estimated τs (represented by cycles) at different color contrast levels for individual participants. (**A** The upper panel plots the CRFs (red) in the L–M axis and (**B** the lower panel the CRFs (blue) in the S–(L + M) axis

Author contributions C.L., J.C. and Z.C. designed research; C.L. performed research; J.C. and Z.C. supervised research; C.L., J.C. and Z.C. analyzed data; C.L., J.C. and Z.C. wrote the article.

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Data availability The data and materials for all experiments are available via this link: https://osf.io/hu7yw/.

Code availability The codes of parameter estimation for all experiments are available via this link: https://osf.io/hu7yw/.

Declarations

Conflicts of interest There is no conflict of interest to disclose.

Ethical approval The procedure for both experiments was approved by the ethics committee of Shanghai University of Sport (No. 102772019RT009).

Consent to participate All the participants signed written informed consent forms in accordance with the Declaration of Helsinki.

Consent for publication All participants provided written consent for publication of anonymized data.

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